

MEMO Test 1

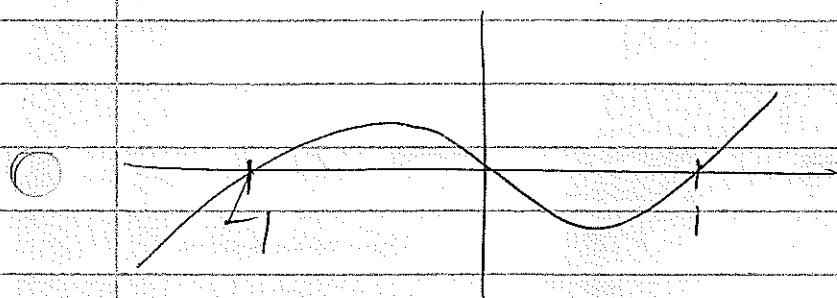
MEMO

1.1 $\frac{x^3 - x}{|x - 1|} > 0$

Since $|x - 1| > 0$

we have: $x^3 - x > 0$

$$\Rightarrow x(x^2 - 1) > 0$$



and $x \in (-1, 0) \cup (1, \infty)$

1.2 $-2 \leq \frac{1}{x} < -1$
 $x < 0$

$$-2 \leq \frac{1}{x}$$

$$-2x \geq 1$$

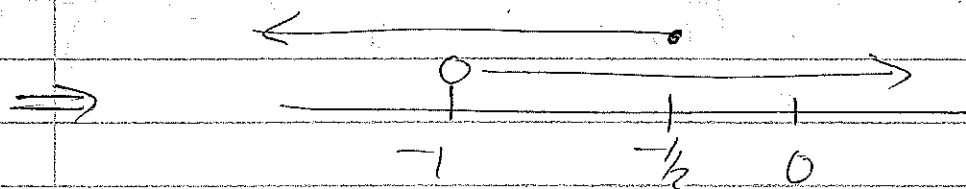
$$x \leq -\frac{1}{2}$$

$$\text{and } \frac{1}{x} < -1$$

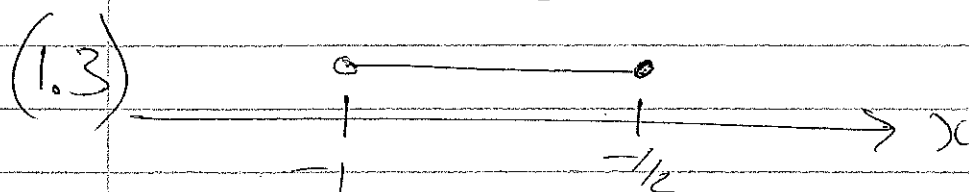
$$\text{and } 1 > -x$$

$$\Rightarrow -1 < x$$

$$\Rightarrow x > -1$$

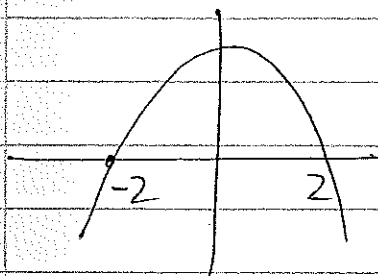


$$\Rightarrow x \in (-1, -\frac{1}{2}]$$



$$(1.4) \quad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{\sqrt{4-x^2}}{x-2}$$

$$4-x^2 \geq 0$$



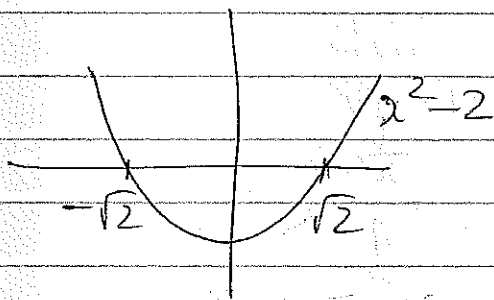
$$\Rightarrow x \in [-2, 2]$$

$$\text{But } x \neq 2$$

$$\Rightarrow \underline{x \in [-2, 2)}$$

$$D_{f/g} = [-2, 2)$$

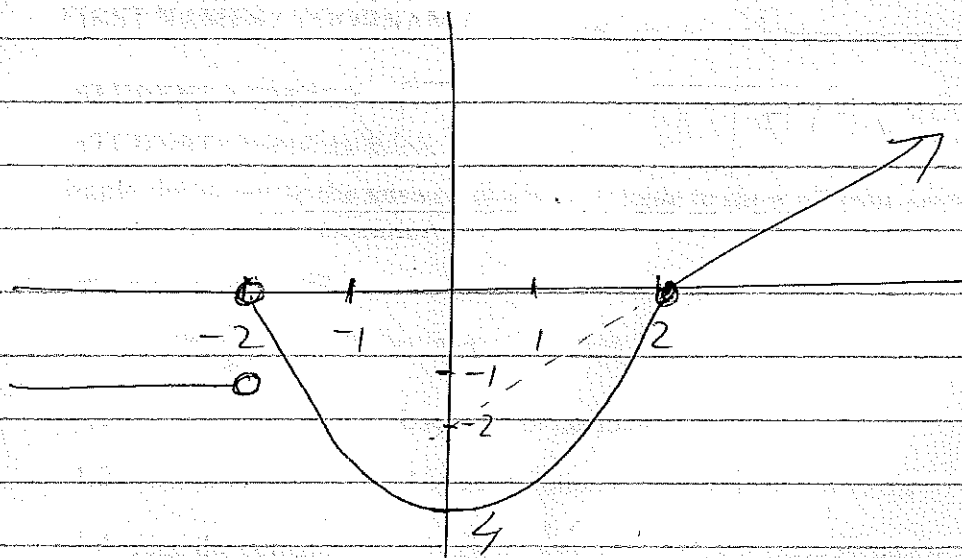
$$(1.5) \quad |x^2-2| = \begin{cases} x^2-2 & \text{if } x^2-2 \geq 0 \\ -(x^2-2) & \text{if } x^2-2 < 0 \end{cases}$$



$$\Rightarrow |x^2-2| = \begin{cases} x^2-2 & \text{if } x \in (-\infty, -\sqrt{2}] \cup [\sqrt{2}, \infty) \\ -x^2+2 & \text{if } x \in (-\sqrt{2}, \sqrt{2}) \end{cases}$$

Question 2:

$$f(x) = \begin{cases} -1 & \text{if } x < -2 \\ x^2 - 4 & \text{if } |x| < 2 \\ x - 2 & \text{if } x \geq 2 \end{cases}$$



$$D_f = (-\infty, -2) \cup (-2, \infty)$$

$$R_f = [-4, \infty)$$

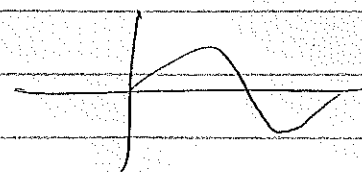
3.1. $\operatorname{cosec} x = \operatorname{cosec} \frac{15\pi}{6}$

$$= \frac{1}{\sin\left(\frac{12\pi + 3\pi}{6}\right)}$$

$$= \frac{1}{\sin(2\pi + \frac{\pi}{2})}$$

$$= \frac{1}{\sin \frac{\pi}{2}}$$

$$= 1$$

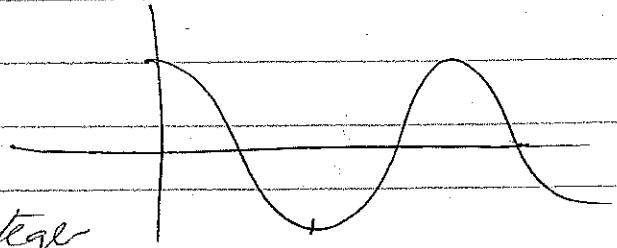


(3.2)

$$\cos \theta = -1$$

$$\Rightarrow \theta = \pi + 2k\pi$$

$k \text{ any integer}$



$$\Rightarrow \theta = (2k+1)\pi, k \text{ any integer}$$
